

Sharmila S

Data Science Student

+91 93606 80762 | ssharmi1509@gmail.com | [LinkedIn](#) | [GitHub](#)

Professional Summary

MCA Candidate with a focus on building intelligent, **automated systems** that **solve real-world problems**. I specialize in architecting **Agentic AI** workflows using **CrewAI**. Most recently, I developed an **XGBoost**-driven engine that optimized water demand forecasting and automated "Smart Pumping" logic, successfully **reducing** the campus **carbon footprint** through data-driven energy management.

Technical Skills

- **Artificial Intelligence & Machine Learning**
 - **Modeling:** Supervised Learning, **XGBoost**, Random Forest, Regression Analysis.
 - **Agentic AI:** Multi-agent orchestration with **CrewAI**, NLP Fundamentals.
 - **Strategy:** Model Evaluation, Feature Engineering, **Decision Support Systems**.
- **Data Engineering & Analytics**
 - **Languages:** Python (Pandas, NumPy, Scikit-learn), SQL (MySQL).
 - **Frameworks:** dbt (Familiar), Data Modeling, Big Data Computing.
 - **Visualization:** **Streamlit** (Live Dashboarding), Power BI, UI/UX Design Principles.
- **Software Development & Operations**
 - **Web Stack:** Django, HTML5, CSS3, Git, GitHub.
 - **Systems:** Cloud Computing (**NPTEL Elite Certified**), CI/CD basics.
 - **Environment:** Experience in Linux-based cloud production and model serialization (**Joblib**).

Technical Projects

- **[Ai-Water-Demand-Forecasting](#):** Architected an XGBoost predictive engine to forecast institutional water demand, achieving 94% forecasting accuracy across multi-source data. Designed a "Smart Pumping" DSS that automated scheduling to leverage off-peak electricity rates, reducing campus energy costs by 18%.
- **[Echo Agent-The AI Podcaster](#):** Architected a multi-agent system using **CrewAI** that reduced manual production time by **95%**, automating research, scriptwriting, and voice synthesis. Successfully generated high-fidelity episodes in under **5 minutes**, maintaining consistent narrative flow and 100% automated script accuracy.
- **[Stock Price Prediction Model](#):** Developed a Random Forest Regressor to forecast 5-day trends for HDFC and ICICI Bank, achieving a 92% predictive accuracy rate.

Optimized feature engineering on historical data to reduce model error (MSE) by 15%, providing high-reliability signals for short-term market trends.

Experience

Web Developer Intern - JRMIInfotech, Hosur | January 2025 | On-site

- **Full-Stack Development:** Gained hands-on proficiency in **Python, Django, HTML,** and **CSS** to build and maintain responsive web applications.
- **Database Management:** Developed practical knowledge of **MySQL** for architecting and managing internal data storage systems.
- **Performance Optimization:** Applied backend logic improvements to Python scripts, achieving a **15% reduction in latency** and increasing data processing speeds.
- **Technical Integration:** Bridged the gap between complex backend logic and **UI/UX design principles** to ensure high-performance user interfaces.

SAP Intern - Titan Company Limited, Hosur | August 2024 | On-site

- **Enterprise System Support:** Assisted in identifying and resolving technical tickets to maintain operational efficiency for large-scale enterprise systems.
- **Quality Assurance:** Executed test scripts for new system configurations and reported bugs to ensure stable deployments.
- **Collaborative Training:** Partnered with the technical team to complete assigned assessments and learn standard industry workflows.

NPTEL Certifications

- Cloud Computing (**Elite - 70%**)
- Big Data Computing (**Elite - 69%**)
- Introduction to Internet of Things (**Elite - 69%**)

Education

- MCA(Data Science) - Alliance University ,Bengaluru | 2025 - **Present** | 9.2 SGPA
- B.Sc(CS) - Christ College of Science and Management ,Malur | 2025 | 8.6 CGPA

Leadership and Involvement

Christ College of Science and Management

- **Technical Director** | Women's Empowerment Cell (WEC): Led technical initiatives and managed digital assets from November 2024 to June 2025.
- **Arts and Cultural Secretary** | SACC Association: Managed organizational tasks and coordinated events between October 2024 and June 2025.
- **Peer Counsellor:** Provided support and guidance to fellow students, demonstrating strong interpersonal and communication skills.